## Correlated random effects: `iid1d`, `iid2d`, `iid3d`, `iid4d` and `iid5d`

This model is available for dimensions $p = 1$, 2, 3, 4 and 5. We describe in detail the case for $p = 2$, and then the changes required for $p = 1$, $p = 3$, $p = 4$ and $p = 5$

### Parametrization

The 2-dimensional Normal-Wishard model is used if one want to define two vectors of "random effects", $u$ and $v$, say, for which $(u_i, v_i)$ are iid bivariate Normals

$$\begin{pmatrix} u_i \\ v_i \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \mathbf{W}^{-1}\right)$$

where the covariance matrix $\mathbf{W}^{-1}$ is

$$\mathbf{W}^{-1} = \begin{pmatrix} 1/\tau_a & \rho/\sqrt{\tau_a \tau_b} \\ \rho/\sqrt{\tau_a \tau_b} & 1/\tau_b \end{pmatrix} \tag{1}$$

and $\tau_a$, $\tau_b$ and $\rho$ are the hyperparameters.

Note that $\rho$ is the correlation coefficient, and that $\tau_a$ and $\tau_b$ are the *marginal precisions*, *not* the elements in the precision matrix.

For these models the precision matrix $\mathbf{W}$ is Wishart distributed

$$\mathbf{W} \sim \text{Wishart}_p(r, \mathbf{R}^{-1}), \quad p = 2$$

with density

$$\pi(\mathbf{W}) = c^{-1} |\mathbf{W}|^{(r-(p+1))/2} \exp\left\{-\frac{1}{2}\text{Trace}(\mathbf{W}\mathbf{R})\right\}, \quad r > p + 1$$

and

$$c = 2^{(rp)/2} |\mathbf{R}|^{-r/2} \pi^{(p(p-1))/4} \prod_{j=1}^{p} \Gamma((r + 1 - j)/2).$$

Then,

$$\text{E}(\mathbf{W}) = r\mathbf{R}^{-1}, \quad \text{and} \quad \text{E}(\mathbf{W}^{-1}) = \mathbf{R}/(r - (p + 1)).$$

### Hyperparameters

The hyperparameters are

$$\theta = (\log \tau_a, \log \tau_b, \tilde{\rho})$$

where

$$\rho = 2\frac{\exp(\tilde{\rho})}{\exp(\tilde{\rho}) + 1} - 1$$

The prior-parameters are

$$(r, R_{11}, R_{22}, R_{12})$$

where

$$\mathbf{R} = \begin{pmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{pmatrix}$$

and $r_{12} = R_{21}$ due to symmetry.

The `inla` function reports posterior distribution for the hyperparameters $\tau_a, \tau_b, \rho$ in equation (1).

The prior for $\theta$ is **fixed** to be `wishart2d`

### Specification

The model `iid2d` is specified as

```
y ~ f(i, model="iid2d",n = <length>) + ...
```

and the `iid2d` model is represented internally as one vector of length $n$,

$$(u_1, u_2, \ldots, u_m, v_1, v_2, \ldots, v_m)$$

where $n = 2m$, and $n$ is the (required) argument in `f()`.

For this model the argument `constr=TRUE` is interpreted as

$$\sum u_i = 0, \qquad \text{and} \qquad \sum v_i = 0.$$

### Hyperparameter spesification and default values

**doc** `Gaussian random effect in dim=2 with Wishart prior`

**hyper**

**theta1**

**hyperid** `26001`
**name** `log precision1`
**short.name** `prec1`
**initial** `4`
**fixed** `FALSE`
**prior** `wishart2d`
**param** `4 1 1 0`
**to.theta** `function(x) log(x)`
**from.theta** `function(x) exp(x)`

**theta2**

**hyperid** `26002`
**name** `log precision2`
**short.name** `prec2`
**initial** `4`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log(x)`
**from.theta** `function(x) exp(x)`

**theta3**

**hyperid** `26003`
**name** `logit correlation`
**short.name** `cor`
**initial** `4`
**fixed** `FALSE`

**prior** none
**param**
**to.theta** function(x) log((1 + x) / (1 - x))
**from.theta** function(x) 2 * exp(x) / (1 + exp(x)) - 1

**constr** FALSE

**nrow.ncol** FALSE

**augmented** TRUE

**aug.factor** 1

**aug.constr** 1 2

**n.div.by** 2

**n.required** TRUE

**set.default.values** TRUE

**pdf** iid123d

## Example

In these examples we demonstrate the use of the iid2d-model, with observations that are without noise (essentially).

```
n = 1000
N = 2*n

## need it to simulate data
library(mvtnorm)

if (TRUE)
{
    ## first example - each variable in the correlated pair has its own row in data

    #Using fixed covariance matrix
    rho = 0.5
    ## set variances
    Sigma = matrix(c(1/1, NA, NA, 1/2), 2, 2)
    ## and the correlation
    Sigma[1,2] = Sigma[2,1] = rho*sqrt(Sigma[1,1]*Sigma[2,2])

    y = yy = rmvnorm(n, sigma=Sigma)
    y = c(y[,1], y[,2])

    i = 1:N
    formula = y ~ f(i, model="iid2d", n=N)

    r = inla(formula, data = data.frame(i,y),
            control.family=list(initial=10,fixed=TRUE))
```

```
    print(summary(r))

    print(1/diag(cov(yy)))
    print(cor(yy)[1,2])
}

if (TRUE)
{
    ## second example - both correlated variables occur in the same row of data

    #drawing covariance matrix from hyperprior
    #Let's specify non-default values, expecting strong covariance
    r = 4
    R11 = 5
    R22 = 4
    R12 = 3

    R = matrix(c(R11,R12,R12,R22), 2, 2)
    #Take a single sample from wishart_2(r,R^-1)
    W = rWishart(1,r,solve(R))[,,1]
    Sigma = solve(W) #Compute the covariance matrix


    y = yy = rmvnorm(n, sigma=Sigma)
    z = rnorm(n)
    zz = rnorm(n)
    y = y[,1] + z*y[,2] + zz
    i = 1:n
    j = n + 1:n
    formula = y ~ f(i, model="iid2d", n=N) + f(j,z,copy="i") + zz

    r = inla(formula, data = data.frame(i,j,y,z,zz),
            control.family=list(initial=10,fixed=TRUE),keep=T)
    print(summary(r))

    #The params as in the Sigma matrix
    print(1/diag(Sigma))
    print(cov2cor(Sigma)[1,2])

    #The params as seen in data
    print(1/diag(cov(yy)))
    print(cor(yy)[1,2])
}
```

### The case $p = 1$

For $p = 1$ the hyperparameter is the marginal precision

$$\theta = \log \tau_1$$

The prior is fixed to be `wishart1d` with parameters

$$param = \; r \; R_{11}$$

where

$$\mathbf{R} = \left[ \; R_{11} \; \right]$$

**Hyperparameter spesification and default values**

**doc** `Gaussian random effect in dim=1 with Wishart prior`

**hyper**

**theta**

**hyperid** `25001`
**name** `precision`
**short.name** `prec`
**initial** `4`
**fixed** `FALSE`
**prior** `wishart1d`
**param** `2 1e-04`
**to.theta** `function(x) log(x)`
**from.theta** `function(x) exp(x)`

**constr** `FALSE`

**nrow.ncol** `FALSE`

**augmented** `FALSE`

**aug.factor** `1`

**aug.constr**

**n.div.by**

**n.required** `FALSE`

**set.default.values** `TRUE`

**pdf** `iid123d`

### The case $p = 3$

For $p = 3$ the hyperparameters are

$$\theta = (\log \tau_1, \log \tau_2, \log \tau_3, \tilde{\rho}_{12}, \tilde{\rho}_{13}, \tilde{\rho}_{23})$$

The prior is fixed to be `wishart3d` with parameters

$$param = \; r \; R_{11} \; R_{22} \; R_{33} \; R_{12} \; R_{13} \; R_{23}$$

where

$$\mathbf{R} = \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{12} & R_{22} & R_{23} \\ R_{13} & R_{23} & R_{33} \end{bmatrix}$$

The reported hyperparameters are the marginal precisions $\tau_1$, $\tau_2$ and $\tau_3$ and the correlations $\rho_{12}$, $\rho_{13}$ and $\rho_{23}$.

In this case, the internal representation is given as

$$(u_1, u_2, \ldots, u_m, v_1, v_2, \ldots, v_m, w_1, w_2, \ldots, w_m)$$

where $n = 3m$ is a required argument, and where $(u_i, v_i, w_i)$ are trivariate iid Normal.

### Hyperparameter spesification and default values

**doc** `Gaussian random effect in dim=3 with Wishart prior`

**hyper**

**theta1**

**hyperid** `27001`
**name** `log precision1`
**short.name** `prec1`
**initial** `4`
**fixed** `FALSE`
**prior** `wishart3d`
**param** `7 1 1 1 0 0 0`
**to.theta** `function(x) log(x)`
**from.theta** `function(x) exp(x)`

**theta2**

**hyperid** `27002`
**name** `log precision2`
**short.name** `prec2`
**initial** `4`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log(x)`
**from.theta** `function(x) exp(x)`

**theta3**

**hyperid** `27003`
**name** `log precision3`
**short.name** `prec3`
**initial** `4`
**fixed** `FALSE`
**prior** `none`
**param**

**to.theta** `function(x) log(x)`
**from.theta** `function(x) exp(x)`

**theta4**
**hyperid** `27004`
**name** `logit correlation12`
**short.name** `cor12`
**initial** `0`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log((1 + x) / (1 - x))`
**from.theta** `function(x) 2 * exp(x) / (1 + exp(x)) - 1`

**theta5**
**hyperid** `27005`
**name** `logit correlation13`
**short.name** `cor13`
**initial** `0`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log((1 + x) / (1 - x))`
**from.theta** `function(x) 2 * exp(x) / (1 + exp(x)) - 1`

**theta6**
**hyperid** `27006`
**name** `logit correlation23`
**short.name** `cor23`
**initial** `0`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log((1 + x) / (1 - x))`
**from.theta** `function(x) 2 * exp(x) / (1 + exp(x)) - 1`

**constr** `FALSE`

**nrow.ncol** `FALSE`

**augmented** `TRUE`

**aug.factor** `1`

**aug.constr** `1 2 3`

**n.div.by** `3`

**n.required** `TRUE`

**set.default.values** `TRUE`

**pdf** `iid123d`

### The case $p = 4$

For $p = 4$ the hyperparameters are

$$\theta = (\log \tau_1, \log \tau_2, \log \tau_3, \log \tau_4, \tilde{\rho}_{12}, \tilde{\rho}_{13}, \tilde{\rho}_{14}, \tilde{\rho}_{23}, \tilde{\rho}_{24}, \tilde{\rho}_{34})$$

The prior is fixed to be `wishart4d` with parameters

$$param = \; r \; R_{11} \; R_{22} \; R_{33} \; R_{44} \; R_{12} \; R_{13} \; R_{14} \; R_{23} \; R_{24} \; R_{34}$$

where

$$\mathbf{R} = \begin{bmatrix} R_{11} & R_{12} & R_{13} & R_{14} \\ R_{12} & R_{22} & R_{23} & R_{24} \\ R_{13} & R_{23} & R_{33} & R_{34} \\ R_{14} & R_{24} & R_{34} & R_{44} \end{bmatrix}$$

The reported hyperparameters are the marginal precisions $\tau_1$, $\tau_2$, $\tau_3$ and $\tau_4$, and the correlations $\rho_{12}$, $\rho_{13}$, $\rho_{14}$, $\rho_{23}$, $\rho_{24}$ and $\rho_{34}$.

In this case, the internal representation is given as

$$(u_1, u_2, \ldots, u_m, v_1, v_2, \ldots, v_m, w_1, w_2, \ldots, w_m, x_1, x_2, \ldots, x_m)$$

where $n = 4m$ is a required argument, and where $(u_i, v_i, w_i, x_i)$ are fourvariate iid Normal.

#### Hyperparameter spesification and default values

**doc** `Gaussian random effect in dim=4 with Wishart prior`

**hyper**

**theta1**

**hyperid** `28001`
**name** `log precision1`
**short.name** `prec1`
**initial** `4`
**fixed** `FALSE`
**prior** `wishart4d`
**param** `11 1 1 1 1 0 0 0 0 0 0`
**to.theta** `function(x) log(x)`
**from.theta** `function(x) exp(x)`

**theta2**

**hyperid** `28002`
**name** `log precision2`
**short.name** `prec2`
**initial** `4`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log(x)`

**from.theta** `function(x) exp(x)`

**theta3**

**hyperid** `28003`
**name** `log precision3`
**short.name** `prec3`
**initial** `4`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log(x)`
**from.theta** `function(x) exp(x)`

**theta4**

**hyperid** `28004`
**name** `log precision4`
**short.name** `prec4`
**initial** `4`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log(x)`
**from.theta** `function(x) exp(x)`

**theta5**

**hyperid** `28005`
**name** `logit correlation12`
**short.name** `cor12`
**initial** `0`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log((1 + x) / (1 - x))`
**from.theta** `function(x) 2 * exp(x) / (1 + exp(x)) - 1`

**theta6**

**hyperid** `28006`
**name** `logit correlation13`
**short.name** `cor13`
**initial** `0`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log((1 + x) / (1 - x))`
**from.theta** `function(x) 2 * exp(x) / (1 + exp(x)) - 1`

**theta7**

**hyperid** `28007`
**name** `logit correlation14`

**short.name** cor14
**initial** 0
**fixed** FALSE
**prior** none
**param**
**to.theta** function(x) log((1 + x) / (1 - x))
**from.theta** function(x) 2 * exp(x) / (1 + exp(x)) - 1

**theta8**
**hyperid** 28008
**name** logit correlation23
**short.name** cor23
**initial** 0
**fixed** FALSE
**prior** none
**param**
**to.theta** function(x) log((1 + x) / (1 - x))
**from.theta** function(x) 2 * exp(x) / (1 + exp(x)) - 1

**theta9**
**hyperid** 28009
**name** logit correlation24
**short.name** cor24
**initial** 0
**fixed** FALSE
**prior** none
**param**
**to.theta** function(x) log((1 + x) / (1 - x))
**from.theta** function(x) 2 * exp(x) / (1 + exp(x)) - 1

**theta10**
**hyperid** 28010
**name** logit correlation34
**short.name** cor34
**initial** 0
**fixed** FALSE
**prior** none
**param**
**to.theta** function(x) log((1 + x) / (1 - x))
**from.theta** function(x) 2 * exp(x) / (1 + exp(x)) - 1

**constr** FALSE

**nrow.ncol** FALSE

**augmented** TRUE

**aug.factor** 1

**aug.constr** 1 2 3 4

**n.div.by** `4`

**n.required** `TRUE`

**set.default.values** `TRUE`

**pdf** `iid123d`

### The case $p = 5$

The case $p = 5$ follows by a direct extention of $p = 3$ and $p = 4$, and is therefore not included.

#### Hyperparameter spesification and default values

**doc** `Gaussian random effect in dim=5 with Wishart prior`

**hyper**

- **theta1**
  - **hyperid** `29001`
  - **name** `log precision1`
  - **short.name** `prec1`
  - **initial** `4`
  - **fixed** `FALSE`
  - **prior** `wishart5d`
  - **param** `16 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0`
  - **to.theta** `function(x) log(x)`
  - **from.theta** `function(x) exp(x)`
- **theta2**
  - **hyperid** `29002`
  - **name** `log precision2`
  - **short.name** `prec2`
  - **initial** `4`
  - **fixed** `FALSE`
  - **prior** `none`
  - **param**
  - **to.theta** `function(x) log(x)`
  - **from.theta** `function(x) exp(x)`
- **theta3**
  - **hyperid** `29003`
  - **name** `log precision3`
  - **short.name** `prec3`
  - **initial** `4`
  - **fixed** `FALSE`
  - **prior** `none`
  - **param**
  - **to.theta** `function(x) log(x)`

**from.theta** `function(x) exp(x)`

**theta4**

**hyperid** `29004`
**name** `log precision4`
**short.name** `prec4`
**initial** `4`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log(x)`
**from.theta** `function(x) exp(x)`

**theta5**

**hyperid** `29005`
**name** `log precision5`
**short.name** `prec5`
**initial** `4`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log(x)`
**from.theta** `function(x) exp(x)`

**theta6**

**hyperid** `29006`
**name** `logit correlation12`
**short.name** `cor12`
**initial** `0`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log((1 + x) / (1 - x))`
**from.theta** `function(x) 2 * exp(x) / (1 + exp(x)) - 1`

**theta7**

**hyperid** `29007`
**name** `logit correlation13`
**short.name** `cor13`
**initial** `0`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log((1 + x) / (1 - x))`
**from.theta** `function(x) 2 * exp(x) / (1 + exp(x)) - 1`

**theta8**

**hyperid** `29008`
**name** `logit correlation14`

**short.name** `cor14`
**initial** `0`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log((1 + x) / (1 - x))`
**from.theta** `function(x) 2 * exp(x) / (1 + exp(x)) - 1`

**theta9**
**hyperid** `29009`
**name** `logit correlation15`
**short.name** `cor15`
**initial** `0`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log((1 + x) / (1 - x))`
**from.theta** `function(x) 2 * exp(x) / (1 + exp(x)) - 1`

**theta10**
**hyperid** `29010`
**name** `logit correlation23`
**short.name** `cor23`
**initial** `0`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log((1 + x) / (1 - x))`
**from.theta** `function(x) 2 * exp(x) / (1 + exp(x)) - 1`

**theta11**
**hyperid** `29011`
**name** `logit correlation24`
**short.name** `cor24`
**initial** `0`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log((1 + x) / (1 - x))`
**from.theta** `function(x) 2 * exp(x) / (1 + exp(x)) - 1`

**theta12**
**hyperid** `29012`
**name** `logit correlation25`
**short.name** `cor25`
**initial** `0`
**fixed** `FALSE`
**prior** `none`

**param**
**to.theta** `function(x) log((1 + x) / (1 - x))`
**from.theta** `function(x) 2 * exp(x) / (1 + exp(x)) - 1`

**theta13**
**hyperid** `29013`
**name** `logit correlation34`
**short.name** `cor34`
**initial** `0`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log((1 + x) / (1 - x))`
**from.theta** `function(x) 2 * exp(x) / (1 + exp(x)) - 1`

**theta14**
**hyperid** `29014`
**name** `logit correlation35`
**short.name** `cor35`
**initial** `0`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log((1 + x) / (1 - x))`
**from.theta** `function(x) 2 * exp(x) / (1 + exp(x)) - 1`

**theta15**
**hyperid** `29015`
**name** `logit correlation45`
**short.name** `cor45`
**initial** `0`
**fixed** `FALSE`
**prior** `none`
**param**
**to.theta** `function(x) log((1 + x) / (1 - x))`
**from.theta** `function(x) 2 * exp(x) / (1 + exp(x)) - 1`

**constr** `FALSE`

**nrow.ncol** `FALSE`

**augmented** `TRUE`

**aug.factor** `1`

**aug.constr** `1 2 3 4 5`

**n.div.by** `5`

**n.required** `TRUE`

**set.default.values** `TRUE`

**pdf** `iid123d`

### Notes

The model `iid1d` is similar to the model `iid` (and included for completeness only). The prior for `iid1d` is fixed to be Wishart-distributed, which reduces to a Gamma-distribution for the precision with parameters

$$a = r/2 \qquad \text{and} \qquad b = R_{11}/2$$

hence

```
y ~ f(i, model="iid1d", hyper = list(theta=list(param=c(3, 4))))
```

is equivalent to

```
y ~ f(i, model="iid", hyper = list(theta=list(param=c(1.5, 2), prior="loggamma")))
```